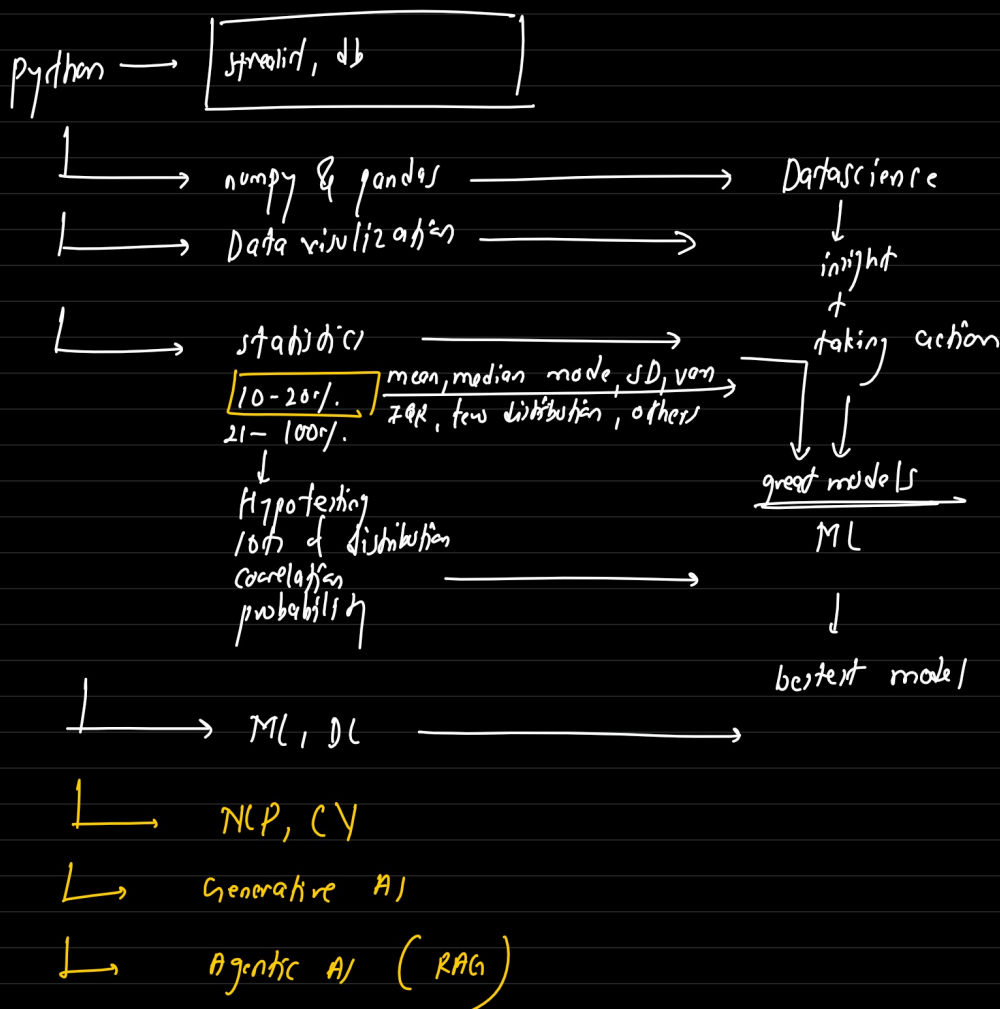


04-09-2026

Agenda:

statistics - 3



Recap:

mean, median, mode → measurement of central tendency

variance, SD → ——— dispersion

Formula:

variance: (σ^2)

population mean

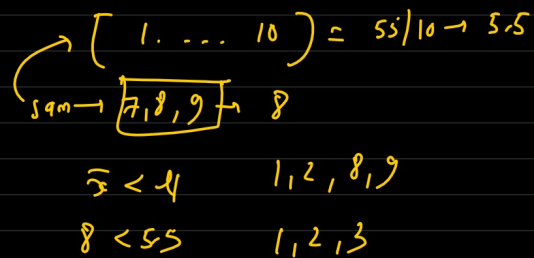
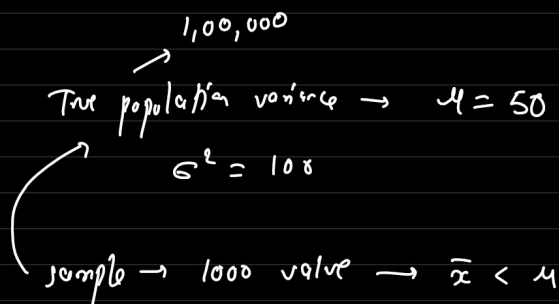
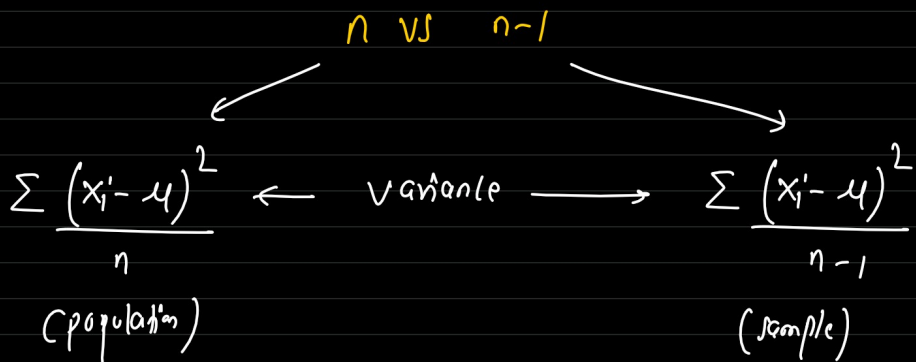
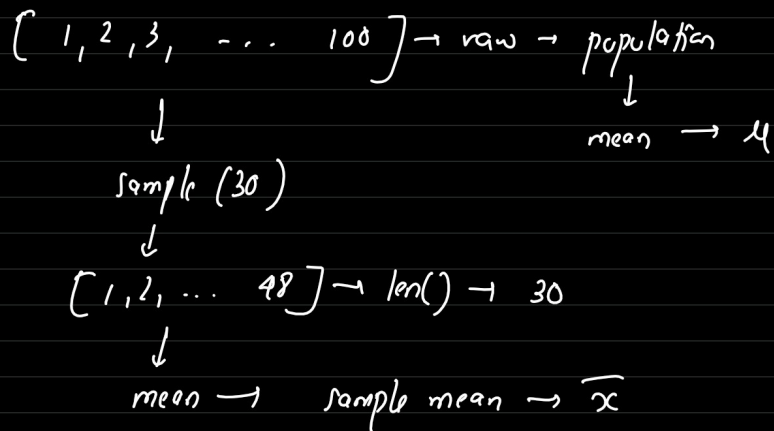
$$\sum \frac{(x_i' - \mu)^2}{n}$$

sample mean

$$\sum \frac{(x_i' - \bar{x})^2}{n-1}$$

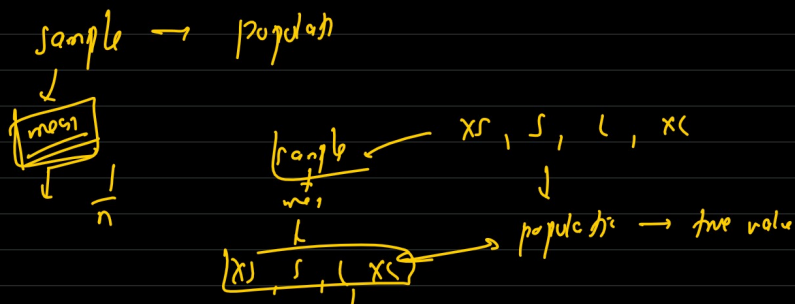
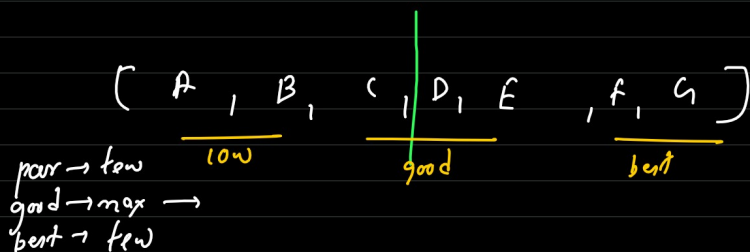
↗

population



$\mu = 30$

$1000 \rightarrow \boxed{x < \mu}$

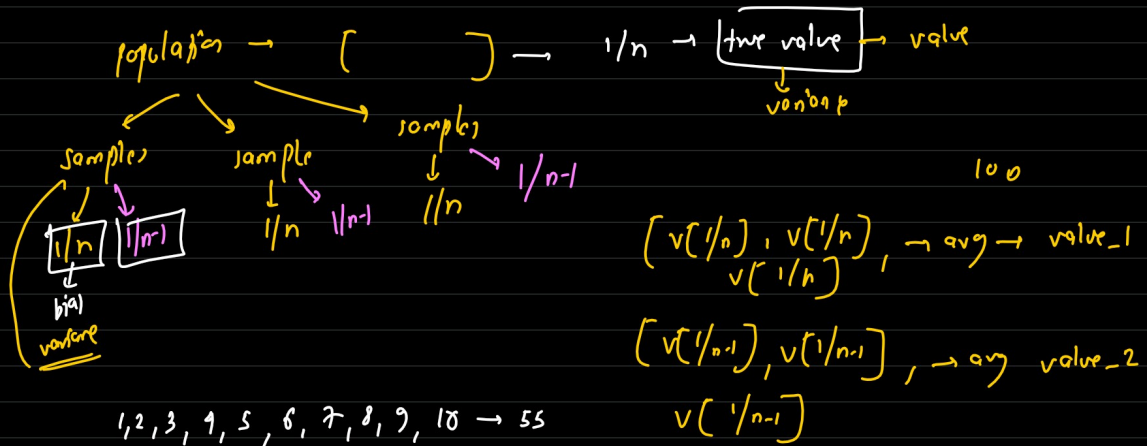


population $\rightarrow$ mean $\rightarrow$ correct value

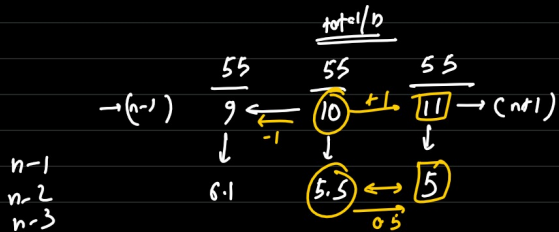
sample $\rightarrow$ moon $\rightarrow$ underestimated value (problem)

✓ $\bar{x} < \mu$
bias

Simulation $\rightarrow$ replica of original experiment

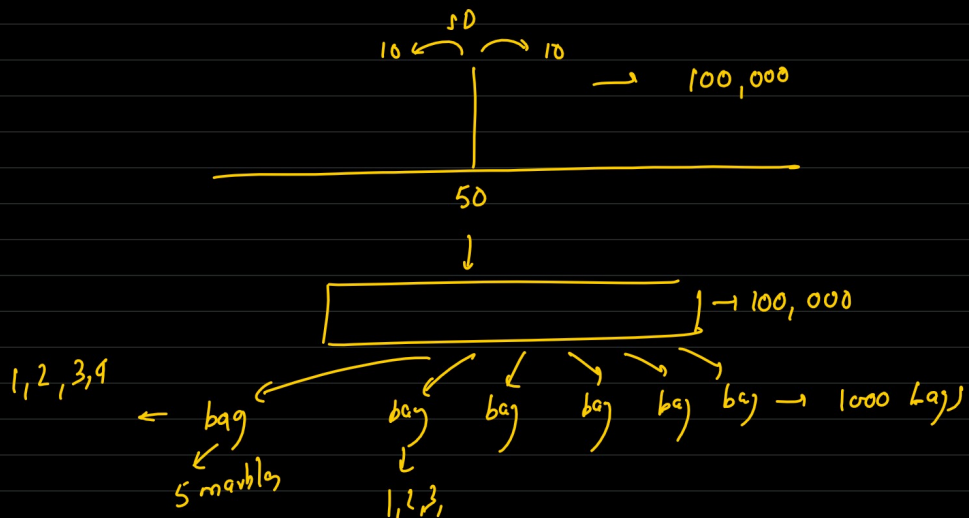
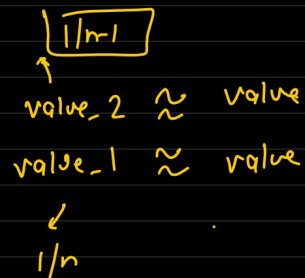


1, 2, 3, 4, 5, 6, 7, 8, 9, 10 $\rightarrow$ 55


$$\begin{array}{l} n-1 \\ n-2 \\ n-3 \end{array}$$

$\bar{x} < \mu \rightarrow$ increase the $\bar{x}$
to have the
correction

$\bar{x} = 14$
↑
estimate
of true population



Divide by $n-1$ instead of n , inflates the result just enough to cancel out the underestimate. $\rightarrow$ Bessel's correction

$$1/n \quad 1/n-1$$

population $\rightarrow$ mean is not something you derive from your data
 $\rightarrow$ mean is fixed property of the population.

$\left[\right] \rightarrow$ mean is external to the dataset.
 $\mu:$
 $\rightarrow 100,000$
 $\uparrow$

country, data $\rightarrow$ all 18 year old in country. $x \rightarrow \mu: 165 \text{ cm}$

variance of 5 people

$$\begin{aligned} x_1 - \mu &= \text{var} \\ x_2 - \mu &= \text{var} \\ x_3 - \mu &= \text{var} \\ x_4 - \mu &= \text{var} \\ x_5 - \mu &= \text{var} \\ &\downarrow \\ &\text{external} \end{aligned}$$

5 people
 $\textcircled{163}$ $\rightarrow$ intern
 ext

5 person $\rightarrow 163$

$$\frac{x_1 + x_2 + x_3 + x_4 + x_5}{n=5} = \textcircled{163}$$

$$x_1 + x_2 + x_3 + x_4 + x_5 = 815$$

$$\frac{200 + 100 + 300 + 100 + x_5}{5} = \boxed{163}$$

$$\begin{aligned} 700 + \textcircled{x_5} &= 815 \\ &\swarrow \quad \searrow \\ x_1 + x_2 + x_3 + x_4 &\quad \text{forced value} \end{aligned}$$

degree of free dom

$$\begin{aligned} 4 \text{ free} &= \boxed{815} \\ 1 \text{ non-free} & \end{aligned}$$

$$df = 1$$

$$\underline{n-1} = \textcircled{df}$$

x_i	$\bar{x}$	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
34	47	<u>$34 - 47 = -13$</u>	
47	47	<u>0</u>	
52	47	<u>5</u>	
61	47	<u>14</u>	
41	47	<u>-6</u>	

$$-13 + 0 + 5 + 14 + -6 = 0$$

$$\sum (x_i - \bar{x}) = 0 \quad \text{f choice or assumption}$$

→ fact

fact 1: Degree of freedom (mechanical reason)

$n-1$ value are free → sample

n values are free → population = me

fact 2: Bias correction (reason we care)

percentage & percentile

percentage: ratio expressed out of 100
: what fraction of total it is?

$$\% = \frac{\text{part}}{\text{total}} \times 100 \quad \neq > 100$$

percentile:

what percentage of the group scored below you?

cat → %tile

→ 96 %tile

→ 99 %tile

→ 75 %tile → 75 % of people score less than you

top → exam → 100 %tile → $\frac{50}{100}$

→ $\frac{40}{200}$ $\frac{45}{200}$

one person = $50/100$ → top → 100 %
sec → $49.9/100$

Two main formula to calculate %tile:

12, 18, 24, 30, 35, 41, 47, 55, 63, 70 → sorted

formula 1:

→ school → X → jump

Nearest rank

$$L = \left[\frac{P}{100} \times n \right] \rightarrow \text{ceiling}$$

$P = 30\%$ %tile →

$$L = 0.3 \times 10 = 3$$

3rd value = 24

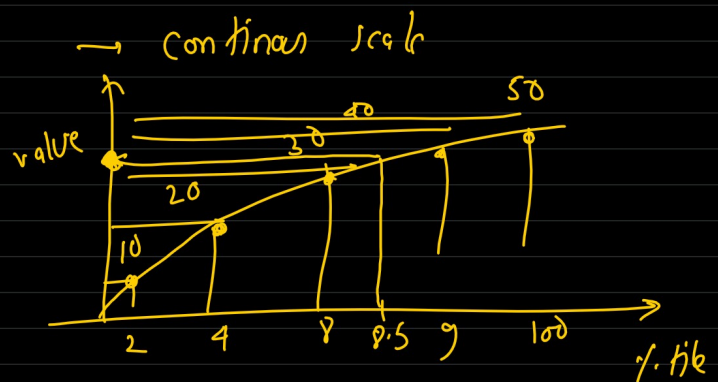
$$p = 32$$

$$L = 0.32 \times 10 = 3.2 \rightarrow \text{ceiling} \rightarrow 4$$

$$p = 39 \rightarrow L \rightarrow 4 \quad 30 > 40$$

Formula 2:
Linear Interpolation (most common)

$$L = \frac{p}{100} \times (n-1) + 1$$



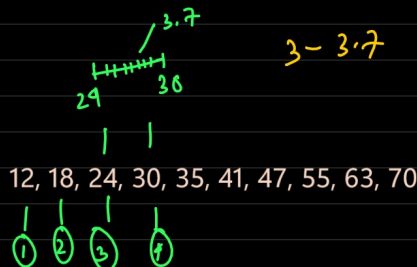
$$p = 30\%$$

$$= \frac{30}{100} \times (10-1) + 1 \rightarrow$$

$$0.3 \times (9) + 1$$

$$\rightarrow 2.7 + 1 \rightarrow 3.7^{\text{th}} \text{ number}$$

BODMAS



$$\text{result} = \text{rank } 3$$

$$\downarrow$$

$$24$$

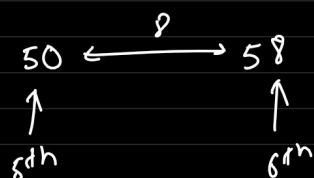
$$\text{rank } 4$$

$$\downarrow$$

$$30$$

$$24 + 0.7 \times 6 \rightarrow 28.2$$

3rd % to go total width



5.9

$$50 + 0.9 \times 8 \rightarrow \underline{\underline{57.2}}$$

index	value
1	29
2	32
3	48
4	49
5	50
6	58
7	62
8	72
9	90
10	100

$$L = \frac{P}{100} \times (n-1) + 1$$

$$\begin{aligned} 100\% &\rightarrow 10 \\ &= \frac{100}{100} \times (9) + 1 \\ &= 9 + 1 \rightarrow 10 \end{aligned}$$

58%

$$\begin{aligned} L &= \frac{58}{100} \times (9) + 1 \\ &= 0.58 \times 9 + 1 \\ &= 5.22 + 1 = \underline{\underline{6.22}} \end{aligned}$$

$$L = 6.22$$

$$L \rightarrow (6) \rightarrow 58$$

$$(.22) \rightarrow ?$$

$$0.22 \times 4$$

$$\rightarrow 0.88$$

$$(6) + (0.22)$$

$$\rightarrow 58 + 0.88$$

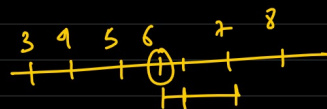
$$\rightarrow \underline{\underline{58.88}} \rightarrow 58\% \text{ tile}$$

$$\downarrow \text{value} \rightarrow 59\% \text{ tile}$$

$$7 - 6$$



$$62 - 58 \rightarrow 4$$

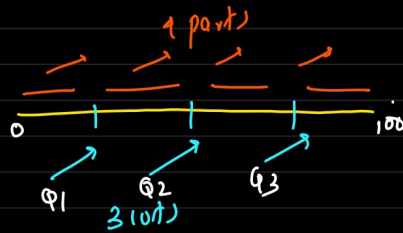


Data visualization $\rightarrow$ Box blot $\rightarrow$ 

Quartiles, IQR, outlier detection:

$\therefore$ file $\rightarrow$ known

Quartiles $\rightarrow$ $\therefore$ files that split data into four equal parts



Min $\rightarrow$

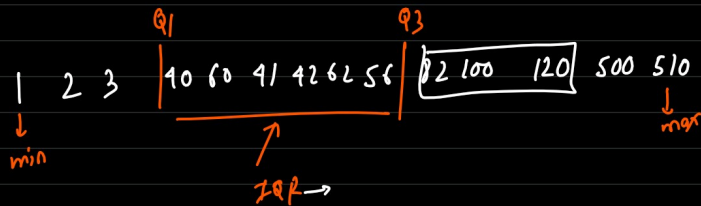
$Q_1 \rightarrow 25^{th}$ $\therefore$ file

$Q_2 \rightarrow 50^{th}$ $\therefore$ file

$Q_3 \rightarrow 75^{th}$ $\therefore$ file

Max $\rightarrow$

IQR $\rightarrow Q_3 - Q_1 \rightarrow$ place/region where bulk of (Inter quartile range) data lives



IQR $\rightarrow ?$

SD $\rightarrow$ spread $\rightarrow$

S1	$\rightarrow$	2
S2	$\rightarrow$	56
S3	$\rightarrow$	50
:	$\rightarrow$	50
S100	$\rightarrow$	98

IQR $\rightarrow 50\%$ $\rightarrow$ 25% file

98 $\rightarrow$ 50
2 $\rightarrow$ 1 $\rightarrow$ 2 SD